

Practice Sheet: SORT and SORTBY Functions

This practice sheet helps you master the **SORT** and **SORTBY** functions in Excel.

These functions are used to arrange data in ascending or descending order. In real offices, sorting is used to arrange employee records by name, salary, department, joining year, performance rating, and many other fields.

This module uses a new employee dataset so that you can practise entering fresh data and then sorting it in different ways.

New Functions: SORT and SORTBY

=SORT(array, [sort_index], [sort_order], [by_col])

Meaning:

- **array:** The full range of data that you want to sort.
- **sort_index:** The column number inside the selected range by which sorting will be done.
- **sort_order:** Use **1** for ascending order and **-1** for descending order.
- **by_col:** Usually left blank because we normally sort rows.

=SORTBY(array, by_array1, [sort_order1], [by_array2], [sort_order2])

Meaning:

- **array:** The full range of data that you want to sort.
- **by_array1:** The column or range based on which sorting will be done.
- **sort_order1:** Use **1** for ascending order and **-1** for descending order.
- **by_array2:** Optional second sorting condition.

Important Note: The SORT and SORTBY functions are available in newer versions of Excel such as Excel 2021 and Microsoft 365.

Step 1: Set Up the Main Data Table

In **Sheet1**, starting from cell **A1**, enter the following table.

Emp ID	Employee Name	Department	Shift	Exp.	Salary (₹)	Rating
E001	Ankit Sharma	Accounts	Morning	3	22000	Good
E002	Rima Das	Sales	Evening	2	18000	Average
E003	Tanmay Bora	Computer	Morning	5	32000	Excellent
E004	Priyanka Dutta	Front Office	Afternoon	1	14500	Good
E005	Arindam Saikia	Accounts	Evening	4	27000	Excellent
E006	Moumita Roy	Sales	Morning	2	19500	Good
E007	Bijoy Kalita	Computer	Afternoon	6	38000	Excellent

Emp ID	Employee Name	Department	Shift	Exp.	Salary (₹)	Rating
E008	Nisha Sharma	HR	Morning	3	24500	Good
E009	Imran Ali	Marketing	Evening	1	16000	Average
E010	Rakesh Nath	Accounts	Afternoon	7	42000	Excellent
E011	Puja Das	Sales	Morning	3	23000	Good
E012	Akash Baruah	Computer	Evening	2	21000	Average
E013	Manisha Paul	HR	Afternoon	5	30000	Excellent
E014	Rohit Singh	Marketing	Morning	4	27500	Good
E015	Sangeeta Gogoi	Front Office	Evening	1	15000	Average
E016	Debajit Deka	Accounts	Morning	6	36000	Excellent
E017	Kavita Roy	Sales	Afternoon	2	20000	Good
E018	Suman Chetia	Computer	Morning	3	26000	Good
E019	Farhana Begum	HR	Evening	4	28500	Good
E020	Kunal Sharma	Marketing	Afternoon	2	19000	Average
E021	Deepika Kalita	Accounts	Morning	5	31500	Excellent
E022	Rajib Das	Sales	Evening	6	35000	Excellent
E023	Neha Kapoor	Computer	Afternoon	1	17000	Average
E024	Hiren Bora	Front Office	Morning	3	21500	Good
E025	Alisha Ahmed	HR	Evening	2	20500	Good
E026	Pranjal Saikia	Marketing	Morning	5	33000	Excellent
E027	Karishma Dey	Accounts	Afternoon	1	15500	Average
E028	Vikram Sinha	Sales	Morning	4	29000	Good
E029	Meera Nair	Computer	Evening	7	45000	Excellent
E030	Abhishek Roy	HR	Afternoon	3	24000	Good
E031	Sneha Dutta	Marketing	Evening	2	18500	Average
E032	Rahul Choudhury	Accounts	Morning	8	48000	Excellent
E033	Ankita Mahanta	Sales	Afternoon	3	23500	Good
E034	Nilotpall Das	Computer	Morning	4	31000	Excellent
E035	Jyoti Devi	Front Office	Evening	2	17500	Average
E036	Manash Sarma	HR	Morning	6	37000	Excellent

Emp ID	Employee Name	Department	Shift	Exp.	Salary (₹)	Rating
E037	Rituparna Bora	Marketing	Afternoon	3	22500	Good
E038	Wasim Ahmed	Accounts	Evening	2	20500	Good
E039	Pankaj Dutta	Sales	Morning	5	34000	Excellent
E040	Sonali Gogoi	Computer	Afternoon	2	23000	Good
E041	Chandan Roy	Front Office	Morning	4	26000	Good
E042	Bipasha Das	HR	Evening	1	16500	Average
E043	Nirmal Kalita	Marketing	Morning	6	39000	Excellent
E044	Aditi Sharma	Accounts	Afternoon	3	25500	Good
E045	Rajashree Deka	Sales	Evening	2	19500	Average
E046	Partha Pratim	Computer	Morning	8	52000	Excellent
E047	Mitali Saikia	HR	Afternoon	4	29500	Good
E048	Saurav Bora	Marketing	Evening	3	24500	Good
E049	Monika Paul	Front Office	Morning	1	14000	Average
E050	Diganta Gogoi	Accounts	Evening	5	32500	Excellent

Step 2: Create Sheet2 for Sorted Results

Create a new sheet named **Sorted Results**.

All SORT and SORTBY formulas should be entered in the **Sorted Results** sheet. This keeps the original employee data safe in Sheet1.

Step 3: Sort by Employee Name from A to Z

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
A1	Sort by Employee Name A to Z
A2	=SORT(Sheet1!A2:G51,2,1)

This sorts the data by the **Employee Name** column in ascending order.

Step 4: Sort by Employee Name from Z to A

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
I1	Sort by Employee Name Z to A
I2	=SORT(Sheet1!A2:G51,2,-1)

This sorts the data by the **Employee Name** column in descending order.

Step 5: Sort by Salary from Lowest to Highest

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
Q1	Sort by Salary Low to High
Q2	=SORT(Sheet1!A2:G51,6,1)

This sorts the data by **Salary** from lowest to highest.

Step 6: Sort by Salary from Highest to Lowest

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
A60	Sort by Salary High to Low
A61	=SORT(Sheet1!A2:G51,6,-1)

This sorts the data by **Salary** from highest to lowest.

Step 7: Sort by Experience from Lowest to Highest

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
I60	Sort by Experience Low to High
I61	=SORT(Sheet1!A2:G51,5,1)

This sorts the data by **Experience** from lowest to highest.

Step 8: Sort by Experience from Highest to Lowest

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
Q60	Sort by Experience High to Low
Q61	=SORT(Sheet1!A2:G51,5,-1)

This sorts the data by **Experience** from highest to lowest.

Step 9: Sort by Department from A to Z

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
A120	Sort by Department A to Z
A121	=SORT(Sheet1!A2:G51,3,1)

This sorts the data by the **Department** column.

Step 10: Sort by Shift from A to Z

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
I120	Sort by Shift A to Z
I121	=SORT(Sheet1!A2:G51,4,1)

This sorts the data by the **Shift** column.

Step 11: Sort by Rating from A to Z

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
Q120	Sort by Rating A to Z
Q121	=SORT(Sheet1!A2:G51,7,1)

This sorts the data by the **Rating** column.

Step 12: SORTBY by Salary Highest to Lowest

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
A180	SORTBY Salary High to Low
A181	=SORTBY(Sheet1!A2:G51,Sheet1!F2:F51,-1)

This uses **SORTBY** to sort the full table based on the **Salary** column.

Step 13: SORTBY by Experience Highest to Lowest

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
I180	SORTBY Experience High to Low
I181	=SORTBY(Sheet1!A2:G51,Sheet1!E2:E51,-1)

This sorts the full table based on the **Experience** column.

Step 14: SORTBY by Department and Salary

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
Q180	SORTBY Department and Salary
Q181	=SORTBY(Sheet1!A2:G51,Sheet1!C2:C51,1,Sheet1!F2:F51,-1)

This sorts the records by **Department** first, and then sorts the employees of each department by **Salary** from highest to lowest.

Step 15: SORTBY by Shift and Employee Name

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
A240	SORTBY Shift and Name
A241	=SORTBY(Sheet1!A2:G51,Sheet1!D2:D51,1,Sheet1!B2:B51,1)

This sorts the records by **Shift** first, and then by **Employee Name** from A to Z.

Step 16: SORTBY by Rating and Salary

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
I240	SORTBY Rating and Salary
I241	=SORTBY(Sheet1!A2:G51,Sheet1!G2:G51,1,Sheet1!F2:F51,-1)

This sorts the records by **Rating** first, and then by **Salary** from highest to lowest.

Step 17: Sort Only Selected Columns

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
Q240	Sort Selected Columns by Salary
Q241	=SORT(Sheet1!A2:F51,6,-1)

This sorts only the columns from **Emp ID** to **Salary** by salary from highest to lowest.

Step 18: Sort Filtered Accounts Employees by Salary

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
A300	Accounts Employees Sorted by Salary
A301	=SORT(FILTER(Sheet1!A2:G51,Sheet1!C2:C51="Accounts"),6,-1)

This first filters only the **Accounts** employees, then sorts them by **Salary** from highest to lowest.

Step 19: Sort Filtered Excellent Employees by Salary

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
I300	Excellent Employees Sorted by Salary
I301	=SORT(FILTER(Sheet1!A2:G51,Sheet1!G2:G51="Excellent"),6,-1)

This first filters only employees with **Excellent** rating, then sorts them by **Salary** from highest to lowest.

Step 20: Sort Filtered Low Salary Employees

In the **Sorted Results** sheet, enter the following:

Cell	Value to Enter
Q300	Low Salary Employees Sorted
Q301	=SORT(FILTER(Sheet1!A2:G51,Sheet1!F2:F51<20000),6,1)

This first filters employees whose salary is below ₹20,000, then sorts them by salary from lowest to highest.

Practice Questions

Use the **SORT** and **SORTBY** functions to solve the following questions. Use the exact cells given below in the **Sorted Results** sheet so that everyone's workbook remains uniform.

Cell	Task
A360	Sort the full table by Emp ID from A to Z.
A420	Sort the full table by Emp ID from Z to A.
A480	Sort the full table by Employee Name from A to Z.
A540	Sort the full table by Employee Name from Z to A.
A600	Sort the full table by Department from A to Z.
A660	Sort the full table by Shift from A to Z.
A720	Sort the full table by Experience from lowest to highest.
A780	Sort the full table by Experience from highest to lowest.

Cell	Task
A840	Sort the full table by Salary from lowest to highest.
A900	Sort the full table by Salary from highest to lowest.
A960	Sort the full table by Rating from A to Z.
A1020	Use SORTBY to sort by Salary from highest to lowest.
A1080	Use SORTBY to sort by Experience from highest to lowest.
A1140	Use SORTBY to sort first by Department , then by Salary from highest to lowest.
A1200	Use SORTBY to sort first by Shift , then by Employee Name from A to Z.
A1260	Use SORTBY to sort first by Rating , then by Salary from highest to lowest.
A1320	Filter only Accounts employees, then sort them by Salary from highest to lowest.
A1380	Filter only Sales employees, then sort them by Employee Name from A to Z.
A1440	Filter only Computer employees, then sort them by Salary from highest to lowest.
A1500	Filter only HR employees, then sort them by Experience from highest to lowest.
A1560	Filter only Marketing employees, then sort them by Employee Name from A to Z.
A1620	Filter employees with salary above ₹30,000 , then sort them by Salary from highest to lowest.
A1680	Filter employees with salary below ₹20,000 , then sort them by Salary from lowest to highest.
A1740	Filter employees with Excellent rating, then sort them by Salary from highest to lowest.
A1800	Filter Morning shift employees, then sort them by Employee Name from A to Z.

Extra Challenge

Create a new sheet named **Dynamic Sort**.

In the **Dynamic Sort** sheet, create the following criteria table in cells **A1:B3**:

Cell	Value to Enter
A1	Criteria
B1	Value
A2	Department
B2	Accounts
A3	Minimum Salary
B3	25000

In cell **A5**, type:

Filtered and Sorted Result

In cell **A6**, enter the following formula:

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=SORT(FILTER(Sheet1!A2:G51,(Sheet1!C2:C51=B2)*(Sheet1!F2:F51>=B3)),6,-1)
```

This formula first filters employees according to the selected department and minimum salary, then sorts the result by salary from highest to lowest.

Now change the values in cells **B2** and **B3** and observe how the result changes automatically.

Homework

Create a new sheet named **Homework**. In the Homework sheet, create a new table of **50 products** starting from cell **A1** with the following columns:

Product ID	Product Name	Category	Brand	Stock	Price	Status
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Then use the **SORT**, **SORTBY**, and **FILTER** functions to find the following. Use the exact cells given below in the **Homework** sheet.

Cell	Task
I1	Sort all products by Product Name from A to Z.
I60	Sort all products by Product Name from Z to A.
I120	Sort all products by Category from A to Z.
I180	Sort all products by Brand from A to Z.
I240	Sort all products by Stock from lowest to highest.
I300	Sort all products by Stock from highest to lowest.
I360	Sort all products by Price from lowest to highest.
I420	Sort all products by Price from highest to lowest.
I480	Sort all products by Status from A to Z.
I540	Use SORTBY to sort first by Category , then by Price from highest to lowest.
I600	Use SORTBY to sort first by Brand , then by Product Name from A to Z.

Cell	Task
I660	Use SORTBY to sort first by Status , then by Stock from lowest to highest.
I720	Filter only products from the Electronics category, then sort by Price from highest to lowest.
I780	Filter only products with price above ₹10,000 , then sort by Price from highest to lowest.
I840	Filter only products with low stock, then sort by Stock from lowest to highest.

Save your workbook as **Excel_Practice_Module15.xlsx** and submit it to your instructor.

End of Module 15